# Selective Modernization and Bounded Responsiveness in Japan's Waste-Incineration Fleet: A Facility-Level Panel Study

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**Abstract**

Japan relies heavily on municipal waste incineration, yet a large share of its incineration fleet still does not convert waste heat into electricity. A one-average-fleet view therefore risks treating all plants as though they face one modernization problem, even though non-generators must first enter generation and mature generators must improve conditional performance. Fleet-average studies blur those margins, while generator-only studies miss the extensive margin altogether. Using Ministry of the Environment data for FY2005–FY2024, this paper estimates both within one national facility panel. It first models observed transition into generation among coded facilities first seen without it, then models energy recovery efficiency within a canonical regression frame of operating generators. Within the coded at-risk frame, transition is selective rather than diffuse: older facilities are less likely than 0–10 year facilities to record transition in the next observed year, while larger facilities are more likely to do so. Within the canonical regression frame, efficiency is lower at older plants and higher at larger, more fully utilized ones, while between-facility heterogeneity dominates within-facility movement. A one-average-fleet view therefore understates both the selectivity of entry and the persistence of performance hierarchy. These patterns are descriptive within the paper's linked samples rather than causal estimates of a single modernization mechanism. For municipal fleet planning, non-generators and mature generators should not be managed as one average segment.

**Keywords:** waste incineration; waste-to-energy; Japan; energy recovery; facility panel; transition

## 1 Introduction

Japan operates one of the world's most incineration-dependent municipal waste systems, yet a large share of the fleet still burns waste without generating electricity from the heat it produces (Ministry of the Environment Japan, 2022; Uno, 2015; Tabata & Tsai, 2016). Sectoral planning increasingly treats that gap as part of the waste sector's decarbonization challenge rather than as a narrow engineering issue (Yamada et al., 2023). The transition problem is therefore not whether incineration exists, but which facilities move into useful energy recovery and what performance looks like once they do.

That pattern is easy to flatten into one average fleet story. But the relevant questions are not the same. One is extensive: which facilities record observed transition into power generation at all? The other is intensive: among plants that already generate, which facilities achieve high energy recovery efficiency? An average-fleet view can therefore make the system look uniformly slow, when the real pattern is selective entry at one end of the fleet and persistent performance hierarchy at the other.

This paper estimates both margins in one national facility-level panel. Its contribution is not simply that it studies Japan, but that it does what fleet-average studies and generator-only studies usually do not: it uses one linked dataset to estimate observed transition into generation and conditional performance within generation side by side. Fleet-average studies blur entry and performance, while generator-only studies miss who enters generation at all. Within the coded at-risk frame, observed transition is selective toward younger and larger facilities. Within the canonical regression frame, efficiency is strongly structured by age, scale, and utilization, while within-facility movement remains limited relative to between-facility heterogeneity.

The gap addressed here is narrower than a generic claim that Japan has been understudied. Waste-to-energy research can describe fleet trajectories, and generator-only studies can explain conditional performance once plants already operate as generators. What remains uncommon is one linked municipal-fleet analysis that estimates both margins and asks whether they point to the same modernization bottleneck. Japan makes that contrast especially visible because a substantial non-generating segment remains active alongside a smaller modern generating segment. The broader claim is correspondingly narrow: in fleets that mix non-generators and mature generators, adoption and conditional performance should be modeled separately before they are interpreted together.

The rest of the paper proceeds as follows. Section 2 positions the paper against the literature most relevant to the analytical split. Section 3 introduces the two linked analytical frames and the main estimation choices. Section 4 reports the adoption and efficiency results in sequence, then ties them together. Section 5 interprets the combined finding, explains what the data still cannot identify, and states a short set of evidence-consistent implications. Section 6 concludes.

## 2 Literature Positioning

This paper speaks to three overlapping literatures: waste-to-energy systems, facility-level efficiency analysis, and infrastructure lock-in. The point is not to survey them exhaustively, because the paper's contribution is narrower than that. The relevant gap is that existing studies often describe fleets in aggregate or explain generator performance conditional on operation, yet rarely estimate both margins in one linked municipal-fleet design.

Work on waste-to-energy systems often documents national trajectories, technology choices, or life-cycle implications of thermal treatment (Astrup et al., 2015; Sun et al., 2018). Japan-specific work has mostly emphasized technology upgrading, heat-use constraints, and sectoral decarbonization scenarios rather than facility-level transition modeling (Uno, 2015; Tabata & Tsai, 2016; Yamada et al., 2023). That literature is indispensable for understanding why energy recovery matters, but it usually treats the incineration fleet as a sectoral category. It can show whether energy recovery is expanding, whether heat supply remains constrained, or whether the sector matters for net-zero planning, but it is less well suited to distinguishing facilities that enter generation from those that do not.

Facility-level efficiency studies are closer to the present paper, but they typically begin after entry has already occurred. Studies in Taiwan, for example, evaluate the efficiency of operating incinerators by decomposing waste-treatment, electricity-generation, or revenue performance within existing plants (Chen et al., 2012; Yeh, 2020). Recent Chinese plant-level work is likewise highly informative about performance differentials inside the generating segment (Cui et al., 2026), but it does not directly address the modernization margin between non-generating and generating facilities. In that sense, such studies answer the intensive-margin question while leaving the extensive-margin question open.

The lock-in literature adds a different empirical expectation. Infrastructure performance may be shaped by durable design choices, inherited scale, and institutional arrangements rather than by frequent large reversals at mature facilities (Unruh, 2000; Seto et al., 2016). The useful implication here is not that incineration plants must be literally irreversible, but that cross-facility differences may matter more than repeated late-life performance resets. That expectation is only partially visible if entry into generation and performance within generation are never separated.

These literatures overlap in practice, but their empirical starting points are different. Fleet-level work can show whether energy recovery is spreading, generator-only work can show what shapes performance once generation already exists, and lock-in work can motivate why mature performance may remain hierarchical. What remains uncommon is a single facility panel that estimates both margins and tests whether they point to the same modernization bottleneck. This paper contributes that linked design.

## 3 Data and Design

The analysis uses the Ministry of the Environment's General Waste Treatment Survey for FY2005–FY2024 (Ministry of the Environment Japan, 2022). The full panel contains 23,599 facility-year rows. Within that, the coded full-fleet frame contains 19,827 observations across 2,948 facilities with usable identifiers. This paper uses two linked samples because one sample cannot answer both parts of the transition problem.

The first analytical frame is the coded adoption frame. It includes facilities first observed without power generation and follows them until they either record observed transition into generation or remain non-generating in the panel window. After excluding left-censored facilities already generating in their first observed year, the adoption risk set contains 13,770 facility-years across 2,035 facilities, with 141 observed first-adoption events. The main adoption model is a lagged discrete-time logit hazard estimated on 11,717 observations across 1,915 facilities and 140 retained events. Predictors are prior-year age band and prior-year design capacity, with year fixed effects, prefecture fixed effects, and facility-clustered standard errors. This is an observed-transition model, not a complete structural model of all possible modernization pathways.

The second analytical frame is the canonical generator frame. It contains operating facilities with positive throughput and positive power output, after standard cleaning and a bounded efficiency measure. The operating generation sample contains 6,660 rows before identifier and regression cleaning. Of those, 907 rows lack official facility codes and are excluded from the canonical regression frame, which leaves 5,683 observations across 1,016 facilities. The dependent variable is winsorized log electricity generated per tonne processed. Main predictors are facility age, design capacity, capacity utilization, waste heating value, and a grid-emission control. The main specifications are pooled OLS, year fixed effects, random effects, and year fixed effects plus random effects, all used as structured descriptive models rather than clean structural estimates (Wooldridge, 2010). The efficiency results should therefore be read as conditional patterns within the canonical regression frame, not as estimates for the entire generating fleet.

**Table 1:** Linked analytical framework

| Margin | Linked sample | Empirical question | Paper role |
|---|---|---|---|
| Adoption margin | Coded at-risk frame: 13,770 facility-years, 2,035 facilities, 141 observed first-adoption events | Which facilities record observed transition into generation? | Shows whether entry into generation is selective rather than diffuse |
| Efficiency margin | Canonical generator frame: 5,683 observations across 1,016 operating generators | How does performance vary once generation already exists? | Shows whether mature generator performance remains bounded |
| Synthesis | Two linked but non-identical analytical frames | Would one average-fleet view misstate the modernization bottleneck? | Shows why entry and mature performance should not be read as one average process |

Note: the adoption margin is estimated with a lagged discrete-time hazard. The efficiency margin is estimated with descriptive pooled, year-FE, and RE panel specifications.

The two layers belong in one paper because they answer sequential parts of the same modernization problem. The adoption layer identifies which facilities appear to enter the generating regime at all. The efficiency layer identifies whether large gains remain available once facilities are already inside that regime. Without the first layer, the paper would reduce transition to generator performance alone. Without the second, it would say who enters generation but not whether large efficiency gaps remain inside the generating segment. The two samples are linked but non-identical, so they should not be read as one causal pathway. They are instead the extensive-margin gate and the conditional-performance layer of the same modernization problem.

The main identification limits are explicit. In the adoption layer, the paper models observed transition within the coded risk set, not unrestricted fleet-wide modernization. In the efficiency layer, age is closely tied to time and within-facility movement is limited, so the defended interpretation is one of structured conditional association rather than strict causal identification. The paper therefore aims for disciplined inference rather than maximal causal reach. All reported effects should be read as conditional associations within the specified samples, not as estimates of structural causality or policy effects. The paper does not claim that the low within-facility variance ratio resolves all fixed-effects concerns; instead, it uses that variance structure to motivate why a cross-facility descriptive model remains substantively useful for the question at hand.

# 4 Results

## 4.1 Adoption into generation is selective rather than diffuse

The adoption results show a strongly selective transition pattern. In the risk set, annual event rates collapse after age 10 and rise sharply across capacity quartiles. Facilities aged 0–10 years account for 102 first-adoption events, while the three older age bands together account for only 39. By capacity, the largest quartile accounts for 99 first-adoption events, whereas the smallest quartile records only 1.

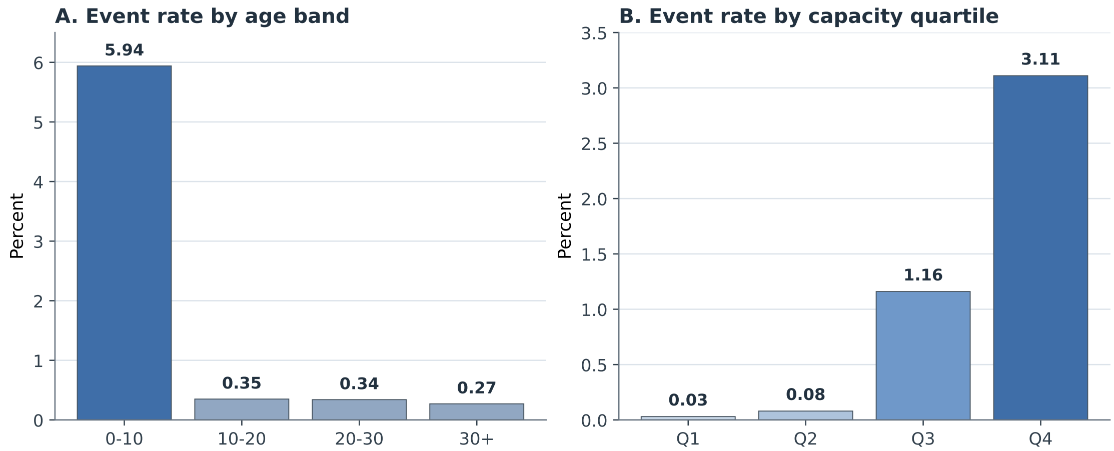


**Figure 1:** Observed adoption event rates by age band and capacity quartile in the coded at-risk frame.

The discrete-time logit hazard compresses that pattern into a clear main result. Relative to 0–10 year facilities, plants aged 10–20 years are about 1.76 percentage points less likely to record transition in the next observed year, plants aged 20–30 years are about 1.72 percentage points less likely, and plants aged 30 years or more are about 1.13 percentage points less likely. Each additional 100 t/day of prior-year design capacity raises annual transition probability by about 0.50 percentage points. The sign pattern is stable in the complementary log-log and linear probability robustness variants.

These effects should be read within the coded at-risk frame, not as a model of all modernization activity in the Japanese fleet. Even within that narrower observed-transition frame, however, the pattern is not one of broad late-life conversion among the old small-plant segment. Observed transition is concentrated among facilities that were already younger and larger before the event year. Older plants do still transition in some cases, but they do so far less often and do not define the main event pattern. The extensive margin therefore looks like selective modernization rather than broad catch-up across the whole fleet.

The pathway audit supports that interpretation without turning it into a stronger mechanism claim than the data earn. Among the 141 observed adoption events, 82 are classified as reset- or rebuild-like, 38 as continuity-type upgrades, 20 as forward-dated or placeholder entries, and 1 as unresolved. This is descriptive pathway evidence, not mechanism identification. The event mix is more consistent with capital-intensive pathways than with diffuse late-life catch-up, but it does not uniquely identify replacement, major refurbishment, or new build as the single pathway. In the main text, the audit therefore functions as a credibility guard rather than as a coequal source of originality.

## 4.2 Performance within generation is bounded and strongly structured

The generator results tell a different but complementary story. Within the canonical regression frame, efficiency is consistently associated with lower values at older facilities and higher values at larger and more fully utilized ones. Across the main specifications, the age coefficient remains negative, the capacity coefficient remains positive, and the utilization coefficient remains positive. The magnitudes differ across models, but the sign pattern is stable, and the emphasis stays on

**Table 2:** Main lagged hazard results for observed transition into generation

| Variable | AME (pp) | SE (pp) |
|---|---|---|
| Prior-year age 10–20 yrs (vs 0–10) | -1.76 | 0.28 |
| Prior-year age 20–30 yrs (vs 0–10) | -1.72 | 0.42 |
| Prior-year age 30+ yrs (vs 0–10) | -1.13 | 0.39 |
| Prior-year capacity (per 100 t/day) | 0.50 | 0.20 |
| *Model summary* | | |
| Observations | | 11,717 |
| Facilities | | 1,915 |
| First-adoption events | | 140 |
| Pseudo-R-squared | | 0.1842 |

Note: entries are average marginal effects in percentage points from the main lagged logit hazard with year and prefecture fixed effects and facility-clustered standard errors.

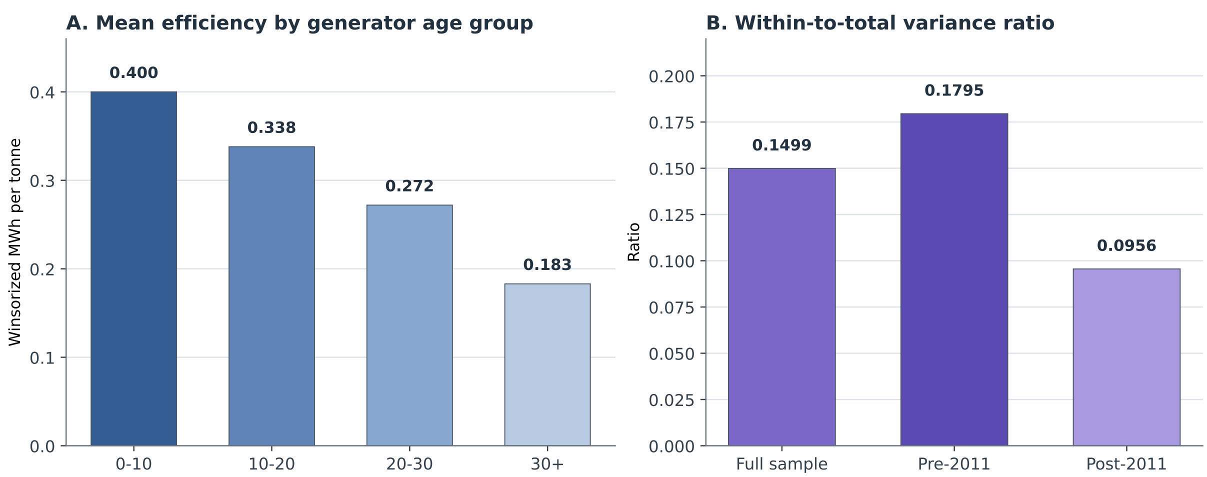


**Figure 2:** Mean efficiency declines across generator age groups, while the within-to-total variance ratio stays low in the full sample and in pre/post-Fukushima splits.

structured conditional association rather than on any single structural parameter.

The strongest descriptive result is not one coefficient but the variance structure. The within-to-total variance ratio of pooled log-efficiency is 0.1499. In other words, the large majority of variation in the dependent variable is between facilities rather than within facilities over time. The ratio remains low in both the pre-Fukushima and post-Fukushima windows, falling from 0.1795 before 2011 to 0.0956 after 2011. That pattern does not prove irreversibility, but it is hard to reconcile with a world in which mature generators frequently undergo large late-life reversals that reshape the fleet distribution. It also does not identify vintage effects separately from all other durable plant characteristics; more narrowly, it supports cross-facility descriptive comparison, not clean causal isolation of vintage itself.

The efficiency margin therefore looks bounded rather than static. Facilities do respond within a design envelope, especially through utilization and operational discipline, but the cross-sectional hierarchy remains strong. Within the canonical regression frame, older facilities are associated with lower efficiency, larger plants are associated with higher efficiency, and utilization matters more as

**Table 3:** Core conditional-efficiency specifications in the canonical generator frame

| Variable | Model 1 Pooled OLS | Model 2 Year FE | Model 3 RE | Model 4 Year FE + RE |
|---|---|---|---|---|
| Facility age (years) | -0.0279*** | -0.0348*** | -0.0188*** | -0.0332*** |
| | (0.0022) | (0.0022) | (0.0025) | (0.0021) |
| Capacity (100 t/day) | 0.0874*** | 0.1030*** | 0.0405*** | 0.0519*** |
| | (0.0083) | (0.0086) | (0.0083) | (0.0096) |
| Capacity utilization | 0.7468*** | 0.7789*** | 0.6199*** | 0.5411*** |
| | (0.1421) | (0.1346) | (0.0997) | (0.0943) |
| Heating value (MJ/kg) | 0.0010 | 0.0032 | 0.0006 | 0.0012 |
| | (0.0023) | (0.0021) | (0.0012) | (0.0010) |
| Grid EF (kg-$CO_2$/kWh) | 0.3182 | -0.4466 | 1.6333*** | -0.1951 |
| | (0.2219) | (0.2714) | (0.1965) | (0.2101) |
| Observations | 5,683 | 5,683 | 5,683 | 5,683 |
| Facilities | 1,016 | 1,016 | 1,016 | 1,016 |
| R-squared | 0.2470 | 0.3721 | 0.1647 | 0.3076 |

Note: standard errors are in parentheses. *** $p < 0.01$. Coefficients are reported as structured conditional associations rather than as strict structural parameters.

a supporting lever within the generating segment than as a fleet-wide equalizer.

This is why the intensive margin cannot be inferred from the adoption margin alone. A facility can be inside generation without being close to the frontier, yet the evidence also suggests that operations alone are unlikely to erase large vintage and scale gaps once plants are mature. Conditional performance is therefore a bounded-performance problem rather than a simple continuation of the entry problem.

### 4.3 Why the two results belong together

Read together, the two margins change the story the fleet appears to tell. The adoption results show that entry into generation is already selective before conditional efficiency is considered, while the efficiency results show that large performance gaps inside the generating segment are not easily erased through within-facility movement alone. A one-average-fleet model would flatten those margins into a single modernization narrative and would therefore understate both the selectivity of entry and the persistence of cross-facility performance differences. The point is not that the two samples form one strict causal chain, but that they identify different constraints within the same fleet. One concerns who gets into generation at all; the other concerns how far operating generators can move once they are already there.

## 5 Discussion

The paper's main interpretive claim is methodological: in this fleet, entry into generation and performance within generation are linked but distinct estimands. Modeling them separately shows that the weakest part of the fleet and the mature generating segment are constrained in different ways. That point matters because municipal fleets often contain both non-generators and mature generators at the same time. If those segments are averaged together, the analyst can see only a muted fleet mean rather than the combination of selective entry and persistent hierarchy that actually structures the system.

The substantive interpretation is correspondingly two-part. On the adoption margin, the data do not support broad late conversion among old small plants. Observed transition within the coded at-risk frame is concentrated among younger and larger facilities, and the pathway audit is more consistent with a capital-intensive event mix than with diffuse late-life catch-up. On the efficiency margin, age, scale, and utilization still matter strongly within the canonical regression frame, while within-facility movement remains modest relative to the cross-sectional hierarchy. Read together, the evidence points more toward selective entry into the generating regime and bounded responsiveness within generation than toward easy convergence once entry has occurred.

That interpretation should remain calibrated. The pathway audit does not prove that replacement is the unique pathway of modernization. The regression results do not provide strict causal estimates of vintage lock-in or clean estimates for all operating generators in Japan. Alternative interpretations remain possible, including reporting compression, unobserved retrofit histories, unmeasured governance differences, and institutional constraints that limit operational responses. The defended claim is therefore narrower: the data support a selective modernization process and a bounded performance envelope, not a uniquely identified mechanism or a full causal hierarchy.

These are evidence-consistent implications, not estimated policy rankings. For the weakest segment, especially older non-generators and small plants, the evidence points more toward capital-renewal planning than toward diffuse late-life operational improvement. For the already-generating segment, utilization, routing, and selective upgrading remain real levers, but they appear more likely to preserve or modestly improve performance within the existing envelope than to eliminate large inherited gaps. For municipal waste planners, the practical takeaway is to separate fleet triage from generator optimization: first identify which non-generators plausibly warrant renewal, then identify which operating generators still have room for incremental gains.

## 6 Conclusion

Japan's incineration transition is not one smooth modernization process. Within the coded adoption frame, observed entry into generation is selective rather than diffuse. Within the canonical generator frame, performance remains stratified by age, scale, and utilization, with limited within-facility movement relative to between-facility differences. Read together, those margins show why a one-average-fleet view can misstate the modernization bottleneck. The paper does not identify one unique pathway or intervention hierarchy, but it does show why municipal fleet studies gain by separating adoption from conditional performance. For readers of municipal waste systems, the practical point is simple: the transition problem at the non-generating end of the fleet is not the same problem as performance management within the already-generating segment, and the two should not be planned as though they were one average task.

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## CRediT Authorship Contribution Statement

Pann Phetra: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Visualization, Writing – original draft, Writing – review & editing.

## Declaration of Competing Interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data Availability

The facility-level source data are derived from the Ministry of the Environment Japan General Waste Treatment Survey. Processed study outputs, manuscript figures, and the associated reproducible analysis workspace can be provided by the author on reasonable request.

## Generative AI and AI-Assisted Technologies Statement

During the preparation of this manuscript, the author used OpenAI Codex and Anthropic Claude to support drafting, language revision, and organizational planning. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the manuscript.

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